# Arjuna JEE 2.0 2027

## Maths

## Basic Mathematics

DPP: 6

## Total Duration - 40 Min

**Q1** Values of $x$ for which $\left(x^2-3\right)(x+1)$ is always negative:

(A) $x \in(-\infty,-\sqrt{3}) \cup(-1, \sqrt{3})$

(B) $x \in(-\sqrt{3}, \sqrt{3})$

(C) $x \in R$

(D) $x \in(-\sqrt{3},-1) \cup(\sqrt{3}, \infty)$

**Q2** Solve $\frac{14 x}{x+1}-\frac{9 x-30}{x-4}<0$

**Q3** Solve $\frac{x^2-5 x+12}{x^2-4 x+5}>3$

**Q4** Solve $\frac{x^2+2}{x^2-1}<-2$

**Q5** Solve $\frac{5-4 x}{3 x^2-x-4}<4$

**Q6** Solve $\frac{(x+2)\left(x^2-2 x+1\right)}{4+3 x-x^2} \geq 0$

**Q7** Solve $\frac{x^4-3 x^3+2 x^2}{x^2-x-30}>0$

**Q8** Solve $\frac{2 x}{x^2-9} \leq \frac{1}{x+2}$

**Q9** Solve $\frac{1}{x-2}+\frac{1}{x-1}>\frac{1}{x}$

**Q10** Solve $\frac{20}{(x-3)(x-4)}+\frac{10}{x-4}+1>0$

**Q11** Solve $\frac{(x-2)(x-4)(x-7)}{(x+1)(x+4)(x+7)}>1$

**Q12** Solve $\left(x^2-2 x\right)\left(2 x-2\right)-9 \frac{2 x-2}{x^2-2 x} \leq 0$

**Q13** If $\log _2\left(4+\log _3(x)\right)=3$, then sum of digits of $x$ is

(A) 3

(B) 6

(C) 9

(D) 18

**Q14** If $\log _5\left(\log _5\left(\log _2 x\right)\right)=0$, then the value of $x$ is

(A) 32

(B) 125

(C) 625

(D) 126

**Q15** If $x^3+8 y^3+x^3 y^3=6 x^2 y^2$ where $x, y \in R-\{0\}$ then value of $-\left(\frac{2}{x}+\frac{1}{y}\right)$ is equal to

# Answer Key

**Q1** (A)

**Q2** $x \in (-1, 1) \cup (4, 6)$

**Q3** $x \in \left(\frac{1}{2}, 3\right)$

**Q4** $x \in (-1, 0) \cup (0, 1)$

**Q5** $x \in \left(-\infty, -\frac{\sqrt{7}}{2}\right) \cup \left(-1, \frac{\sqrt{7}}{2}\right) \cup \left(4/3, \infty\right)$

**Q6** $x \in (-\infty, -2] \cup (-1, 4)$

**Q7** $x \in (-\infty, -5) \cup (1, 2) \cup (6, \infty)$

**Q8** $x \in (-\infty, -3) \cup (-2, 3)$

**Q9** $x \in \left(-\sqrt{2}, 0\right) \cup \left(1, \sqrt{2}\right) \cup \left(2, \infty\right)$

**Q10** $x \in (-\infty, -2) \cup (-1, 3) \cup (4, \infty)$

**Q11** $x \in (-\infty, -7) \cup (-4, -1)$

**Q12** $x \in (-\infty, -1] \cup (0, 1] \cup (2, 3]$

**Q13** (C)

**Q14** (A)

**Q15** 1